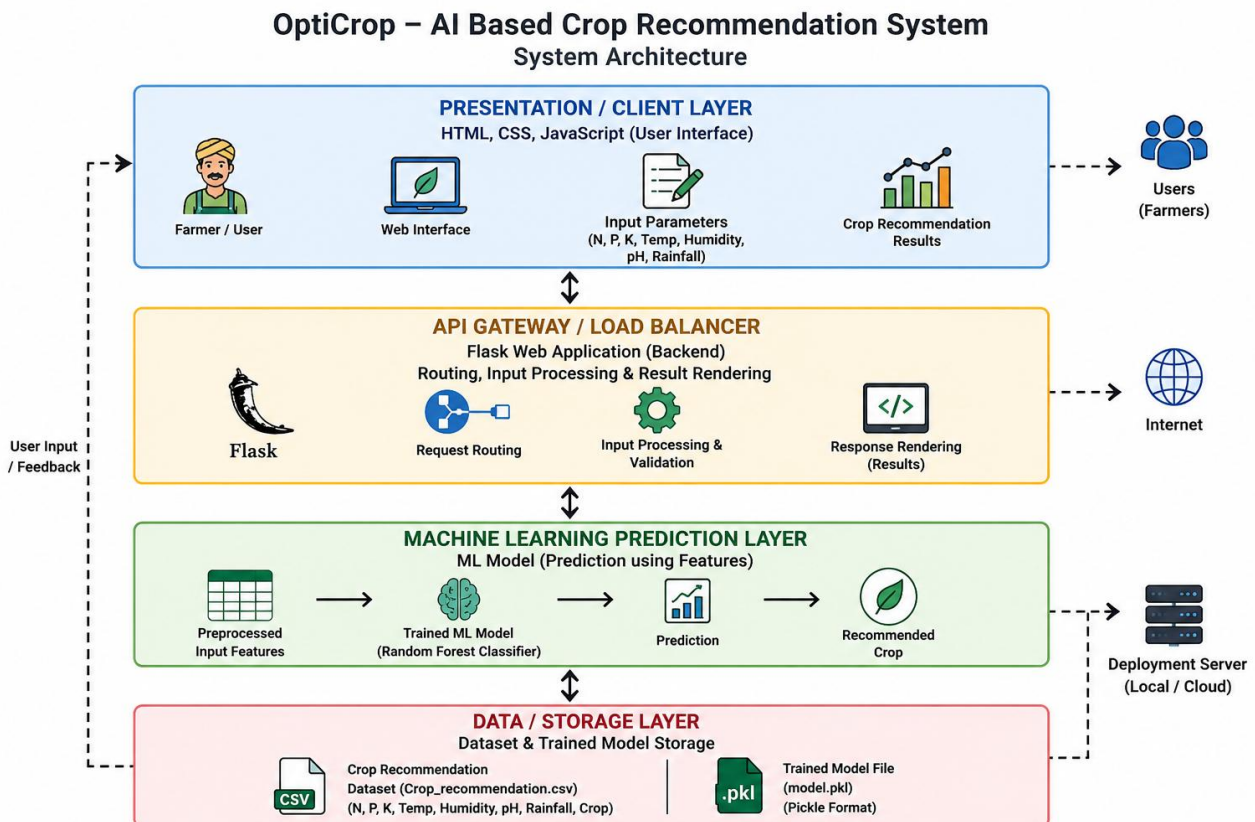


## Solution Architecture

|                      |                                                |
|----------------------|------------------------------------------------|
| <b>Date</b>          | 01 July 2026                                   |
| <b>Team ID</b>       | XXXXXX                                         |
| <b>Project Name</b>  | OptiCrop – AI-Based Crop Recommendation System |
| <b>Maximum Marks</b> | 5 Marks                                        |

### Solution Architecture Diagram:



### Component Description Table

| Component Name         | Description / Role in Architecture                                                                                | Technologies Used    |
|------------------------|-------------------------------------------------------------------------------------------------------------------|----------------------|
| Presentation Layer     | Provides the user interface to collect input values and display crop recommendation results.                      | HTML, CSS            |
| Flask Backend          | Handles routing, processes user input, loads the machine learning model, and returns crop recommendation results. | Python, Flask        |
| Machine Learning Model | Predicts the most suitable crop based on soil nutrients and environmental conditions using trained data.          | Scikit-learn, Pickle |
| Dataset / Storage      | Stores the Crop Recommendation dataset and the trained machine learning model used for prediction.                | CSV, Pickle          |